

PE-EC703C — Wavelet Transforms

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Module 2 Exam Notes: DWT & Multi-resolution Analysis (MRA)

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Contents

1	Multi-resolution Analysis (MRA) Framework	2
1.1	Nested Vector Subspaces	2
1.2	The Five Core MRA Axioms	2
2	Scaling Function and Wavelet Function	3
2.1	The Dilation (Refinement) Equation	3
2.2	Detail Spaces (W_j) and the Wavelet Function	3
3	The Haar Wavelet System	4
3.1	Haar Scaling Function (Father Wavelet)	4
3.2	Haar Wavelet Function (Mother Wavelet)	4
3.3	Haar Wavelet Decomposition Example	4
3.4	Haar Wavelet Packets and Applications	5
3.4.1	Recursive Generation of Haar Wavelet Packets	5
3.4.2	Applications of Haar Wavelet Packets	5

1. Multi-resolution Analysis (MRA) Framework

Definition of Multi-resolution Analysis

Multi-resolution Analysis (MRA) is a mathematical framework developed by Mallat and Meyer to construct orthonormal wavelet bases. It represents a continuous-time signal $x(t) \in L^2(\mathbb{R})$ as a limit of successive approximations, each representing a different resolution level.

1.1 Nested Vector Subspaces

The MRA is characterized by a sequence of closed nested linear vector subspaces $\{V_j\}_{j=-\infty}^{\infty}$ of the Hilbert space $L^2(\mathbb{R})$:

$$\cdots \subset V_{-2} \subset V_{-1} \subset V_0 \subset V_1 \subset V_2 \subset \cdots \subset L^2(\mathbb{R})$$

where:

- Each subspace V_j corresponds to a specific **resolution level** (with larger indices j representing finer details/sampling rates).
- Any signal in V_j is also contained in the next finer subspace V_{j+1} .

1.2 The Five Core MRA Axioms

A family of subspaces $\{V_j\}_{j \in \mathbb{Z}}$ constitutes an MRA of $L^2(\mathbb{R})$ if the following mathematical axioms are satisfied:

1. **Containment (Monotonicity):** $V_j \subset V_{j+1}$ for all $j \in \mathbb{Z}$.
2. **Completeness (Separation):** The intersection of all subspaces contains only the zero function:

$$\bigcap_{j=-\infty}^{\infty} V_j = \{0\}$$

3. **Density:** The union of all subspaces is dense in $L^2(\mathbb{R})$ (any signal can be approximated to arbitrary precision):

$$\overline{\bigcup_{j=-\infty}^{\infty} V_j} = L^2(\mathbb{R})$$

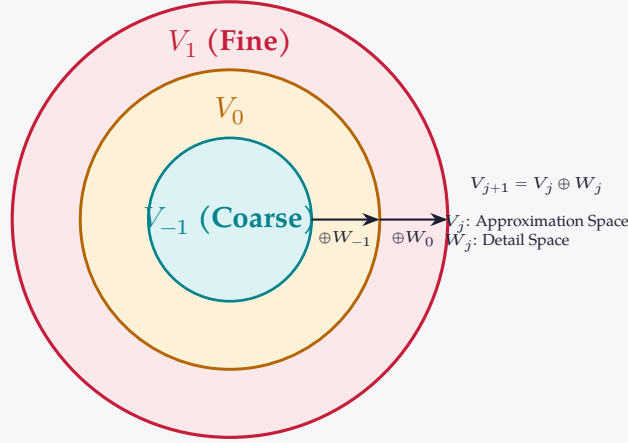
4. **Scaling Similarity (Scale Invariance):** Scaling a function in time by a factor of 2 shifts it to the adjacent subspace:

$$f(t) \in V_j \iff f(2t) \in V_{j+1}$$

5. **Translation Invariance (Shift Invariance):** Shifting a function in V_0 by an integer step remains in V_0 :

$$f(t) \in V_0 \iff f(t - k) \in V_0 \quad \text{for all } k \in \mathbb{Z}$$

Nested Vector Subspaces Hierarchy



2. Scaling Function and Wavelet Function

An MRA is completely defined by its **scaling function** $\phi(t)$ (also known as the **Father Wavelet**).

2.1 The Dilation (Refinement) Equation

Because $\phi(t) \in V_0 \subset V_1$, and the set $\{\sqrt{2}\phi(2t - k)\}_{k \in \mathbb{Z}}$ forms an orthonormal basis for V_1 , the scaling function can be expanded as a linear combination of scaled and shifted copies of itself. This is the **refinement equation** (or **dilation equation**):

$$\phi(t) = \sqrt{2} \sum_{k=-\infty}^{\infty} h_0[k] \phi(2t - k)$$

where $h_0[k]$ are the **scaling filter coefficients** (low-pass filter).

2.2 Detail Spaces (W_j) and the Wavelet Function

The approximation space V_{j+1} has finer resolution than V_j . The missing information between them is represented by the **detail space** W_j , defined as the orthogonal complement of V_j in V_{j+1} :

$$V_{j+1} = V_j \oplus W_j \quad \text{where } V_j \perp W_j$$

By repeating this decomposition:

$$V_{j+1} = V_{j-1} \oplus W_{j-1} \oplus W_j = V_0 \oplus \bigoplus_{i=0}^j W_i$$

To span the detail spaces, we define the **wavelet function** $\psi(t)$ (the **Mother Wavelet**), which can also be written in terms of the scaling function at the next scale:

$$\psi(t) = \sqrt{2} \sum_{k=-\infty}^{\infty} h_1[k] \phi(2t - k)$$

where $h_1[k]$ are the **wavelet filter coefficients** (high-pass filter).

3. The Haar Wavelet System

The **Haar system** is the oldest and simplest example of an MRA.

3.1 Haar Scaling Function (Father Wavelet)

The Haar scaling function $\phi(t)$ is a simple unit step box function:

$$\phi(t) = \begin{cases} 1 & 0 \leq t < 1 \\ 0 & \text{otherwise} \end{cases}$$

Its refinement equation is:

$$\phi(t) = \phi(2t) + \phi(2t - 1)$$

Comparing with the general dilation equation $\phi(t) = \sqrt{2}[h_0[0]\phi(2t) + h_0[1]\phi(2t - 1)]$, the Haar scaling filter coefficients are:

$$h_0[0] = \frac{1}{\sqrt{2}}, \quad h_0[1] = \frac{1}{\sqrt{2}} \quad (\text{all other } h_0[k] = 0)$$

3.2 Haar Wavelet Function (Mother Wavelet)

The Haar wavelet $\psi(t)$ is defined as:

$$\psi(t) = \begin{cases} 1 & 0 \leq t < 0.5 \\ -1 & 0.5 \leq t < 1 \\ 0 & \text{otherwise} \end{cases}$$

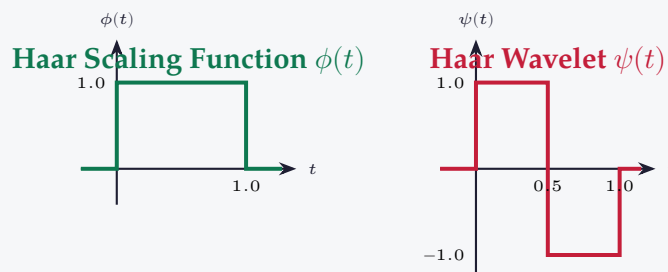
Its refinement equation is:

$$\psi(t) = \phi(2t) - \phi(2t - 1)$$

Thus, the Haar wavelet filter coefficients are:

$$h_1[0] = \frac{1}{\sqrt{2}}, \quad h_1[1] = -\frac{1}{\sqrt{2}} \quad (\text{all other } h_1[k] = 0)$$

Haar Scaling Function and Wavelet Plots



3.3 Haar Wavelet Decomposition Example

Consider a discrete-time signal $x = [c_1, c_2]$ representing values in space V_1 .

1. **Approximation coefficient** (Low-pass projection in V_0):

$$a_1 = \frac{c_1 + c_2}{\sqrt{2}}$$

2. **Detail coefficient** (High-pass projection in W_0):

$$d_1 = \frac{c_1 - c_2}{\sqrt{2}}$$

The Haar transform maps $[c_1, c_2] \rightarrow [a_1, d_1]$. It decomposes the signal into its local mean (coarse trend) and local difference (detail fluctuation).

3.4 Haar Wavelet Packets and Applications

A standard Haar DWT only decomposes the low-pass approximation coefficients (a_j) recursively. A **Haar Wavelet Packet** decomposition recursively filters both the approximation (a_j) and the detail (d_j) coefficients at each level, yielding a full balanced binary tree.

3.4.1 Recursive Generation of Haar Wavelet Packets

Let $W_0(t) = \phi(t)$ be the Haar scaling function and $W_1(t) = \psi(t)$ be the Haar wavelet function. The higher-order wavelet packet functions $W_n(t)$ for $n \geq 0$ are recursively defined by:

$$W_{2n}(t) = \sum_{k=0}^1 h_0[k] W_n(2t - k) = \frac{1}{\sqrt{2}} [W_n(2t) + W_n(2t - 1)]$$

$$W_{2n+1}(t) = \sum_{k=0}^1 h_1[k] W_n(2t - k) = \frac{1}{\sqrt{2}} [W_n(2t) - W_n(2t - 1)]$$

For example, starting from the first level:

- $W_2(t)$ is generated from the low-pass projection of $W_1(t)$, representing a high-frequency approximation packet.
- $W_3(t)$ is generated from the high-pass projection of $W_1(t)$, representing a high-frequency detail packet.

3.4.2 Applications of Haar Wavelet Packets

Due to their simple step-like structure and ease of computational implementation, Haar wavelet packets are widely used in:

- **Transient Signal Detection:** The localized rectangular support of Haar packets makes them highly sensitive to sudden spikes, edges, and discontinuities in signals.
- **High-Frequency Feature Extraction:** By decomposing the high-frequency bands recursively, Haar packets capture structural texture details in images that standard DWT skips.
- **Real-Time Low-Power Hardware Implementations:** Since the Haar filter coefficients ($1/\sqrt{2}$) involve simple additions and subtractions, Haar packet transforms are ideal for low-cost hardware implementation on microcontrollers and FPGAs.

Quick Revision — Module 2

DWT & Multi-resolution Analysis (MRA) | PE-EC703C

Key DWT & MRA Equations

MRA Nested Spaces	$\cdots \subset V_{-1} \subset V_0 \subset V_1 \subset V_2 \cdots \subset L^2(\mathbb{R})$
Scaling Refinement Eq.	$\phi(t) = \sqrt{2} \sum_k h_0[k] \phi(2t - k)$ (Father Wavelet)
Wavelet Refinement Eq.	$\psi(t) = \sqrt{2} \sum_k h_1[k] \phi(2t - k)$ (Mother Wavelet)
Orthogonal Complement	$V_{j+1} = V_j \oplus W_j$ where $V_j \perp W_j$
Haar scaling coefficients	$h_0[0] = 1/\sqrt{2}, \quad h_0[1] = 1/\sqrt{2}$
Haar wavelet coefficients	$h_1[0] = 1/\sqrt{2}, \quad h_1[1] = -1/\sqrt{2}$

One-Line Definitions

- **MRA:** A mathematical framework decomposing a signal into nested resolution spaces.
- **Approximation Space (V_j):** Subspace containing low-resolution signal trends up to scale j .
- **Detail Space (W_j):** Subspace containing the localized high-resolution differences between V_j and V_{j+1} .
- **Father Wavelet ($\phi(t)$):** The scaling function that spans the approximation spaces.
- **Mother Wavelet ($\psi(t)$):** The wavelet function that spans the detail spaces.
- **Dilation Equation:** A recursive relation defining the scaling function using scaled/shifted copies of itself.
- **Haar Wavelet:** The simplest orthonormal wavelet, represented by an asymmetric step function.

Frequently Asked / Viva Questions

Q1. State the five basic axioms of Multi-resolution Analysis (MRA).

The five axioms are: (1) Containment: $V_j \subset V_{j+1}$; (2) Completeness: $\bigcap V_j = \{0\}$; (3) Density: $\overline{\bigcup V_j} = L^2(\mathbb{R})$; (4) Scaling Similarity: $f(t) \in V_j \iff f(2t) \in V_{j+1}$; and (5) Translation Invariance: $f(t) \in V_0 \iff f(t - k) \in V_0$.

Q2. What is the difference between approximation space and detail space?

The approximation space (V_j) represents the low-frequency, smoothed version of a signal at resolution scale j . The detail space (W_j) represents the high-frequency difference or fluctuation needed to transition from a coarse scale V_j to a finer scale V_{j+1} .

Frequently Asked / Viva Questions (contd.)

Q3. Write down the refinement/dilation equation for the scaling function.

The refinement equation is: $\phi(t) = \sqrt{2} \sum_{k=-\infty}^{\infty} h_0[k] \phi(2t - k)$, where $h_0[k]$ are the scaling filter coefficients. This equation shows that the scaling function is self-similar across scales.

Q4. What are the coefficients of the Haar scaling and wavelet filters?

For the Haar scaling filter: $h_0[0] = 1/\sqrt{2}$, $h_0[1] = 1/\sqrt{2}$. For the Haar wavelet filter: $h_1[0] = 1/\sqrt{2}$, $h_1[1] = -1/\sqrt{2}$. All other coefficients are zero.

Q5. Why does the DWT avoid the redundancy of the CWT?

The CWT varies the scale a and shift b continuously, creating infinite overlap and redundant representation. The DWT samples the time-scale grid discretely (usually on a dyadic grid: $a = 2^j, b = k \cdot 2^j$), creating an orthogonal, non-redundant representation.

Q6. What is the relation between the scaling function $\phi(t)$ and the wavelet function $\psi(t)$?

The scaling function $\phi(t)$ generates the approximation spaces (V_j). The wavelet function $\psi(t)$ is orthogonal to $\phi(t)$ and generates the detail spaces (W_j). Mathematically, $\psi(t)$ is derived from the scaling function via: $\psi(t) = \sqrt{2} \sum_k h_1[k] \phi(2t - k)$.

Q7. What are the drawbacks of the Haar wavelet system?

The Haar wavelet is discontinuous in the time domain, which causes poor frequency resolution and spectral leakage. Additionally, it has only one vanishing moment, making it inefficient for compressing smooth signals.

Comparison Table: Approximation vs. Detail Spaces

Feature	Approximation Space (V_j)	Detail Space (W_j)
Spanning Basis	Shifted Scaling Functions: $\{\phi_{j,k}(t)\}$.	Shifted Wavelet Functions: $\{\psi_{j,k}(t)\}$.
Spectral Content	Low-frequency components (trends).	High-frequency components (details/edges).
Filtering Operation	Low-pass filter ($h_0[k]$).	High-pass filter ($h_1[k]$).
Relationship	$V_{j+1} = V_j \oplus W_j$	$W_j \perp V_j$ and $W_j \perp W_i$ (for $i \neq j$).